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REVIEW

The genetics of Crohn's disease and ulcerative colitis – *status quo* and beyond

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Abstract

The two major subtypes of inflammatory bowel disease (IBD), ulcerative colitis (UC, MIM#191390) and Crohn's disease (CD, MIM#266600), are chronic relapsing-remitting inflammatory disorders affecting primarily the gastrointestinal tract. Prevalence rates in North America and Europe range from 21 to 246 per 100,000 for UC and 8 to 214 per 100,000 for CD. Although CD and UC share some clinical and pathological features, they can be distinguished by localization, endoscopic appearance, histology and behavior, which suggest differences in the underlying pathophysiology. The importance of genetic risk factors in disease etiology is high and has been documented more clearly for CD than for UC (relative sibling risks λ_s : 15–35 for CD, 6–9 for UC). The most recent and largest genetic association study for IBD, which employed genome-wide association data for over 75,000 patients and controls, established the association of 163 susceptibility loci with IBD. Although the disease variance explained by the 163 loci only amounts to 13.6% for CD and 7.5% for UC, the identified loci and the candidate genes within yielded valuable insights into the pathogenesis of IBD and the relevant disease pathways. We here review the current research on the genetics of IBD and provide insights into on current efforts as well as suggest topics for future research.

Key Words: *chronic inflammation, Crohn's disease, early-onset IBD, genetic risk, genome-wide association studies, inflammatory bowel disease, risk loci, SNP, ulcerative colitis*

Inflammatory bowel disease as a figurehead for complex diseases: a plethora of genetic disease loci

Inflammatory bowel diseases (IBDs) are chronic inflammatory diseases, which affect different parts of the gastrointestinal tract. Susceptibility to IBD is determined mainly by environmental factors, which remain largely unknown. Genetic susceptibility plays an additional role in disease development and genetic variation can be studied at higher resolution and more accurately than environmental factors. Owing to technology developments, systematic genome-wide and large-scale international studies, the genetic risk map of IBD has grown tremendously. Most identified and robustly replicated loci have been detected by means of genome-wide association studies (GWAS) [1–3]. GWAS are largely hypothesis-free approaches that

identify common risk alleles, which are significantly more frequent in patients compared to healthy individuals. The number of discovered risk loci that are statistically associated with the disease has risen drastically through several independent systematic studies and meta-analyses as summarized in Table I. We are aware of further studies that will report at least 40 additional novel loci, highlighting that mapping efforts continue and that more loci are likely to be identified in the future, in particular, revealing rare variant associations given the growing amounts of whole-exome/-genome resequencing data.

As most Crohn's disease (CD) and ulcerative colitis (UC) patients carry a large number of common risk variants, the effect size of most individual risk variants is low (odds ratio (OR) <1.3; Figure 1). GWAS hence require large case-control sample collections in order to identify the aforementioned small effect sizes and in

Table I. Large-scale association studies performed in IBD until 11/2014. The table likely misses important studies (mostly after 2010) and solely serves to illustrate that a large number of studies have been performed by many international groups and that sample sizes increased over time.

Year	Technology	Author	Disease	Remarks	Country	GWAS cases	GWAS controls	Replication cases	Replication controls
2005	73k	Yamazaki et al. [63]	CD		Japan	94 CD	752	484 CD	345
2006	Illumina 300k	Duerr et al. [64]	CD	Ileal only	North America	567 CD	571	401 CD; 833 trios	433
2007	Nonsynonymous SNP 20k	Hampe et al. [33]	CD		Germany	735 CD	368	498 CD; 380 trios	1032
2007	Affymetrix 100k	Franke et al. [65]	CD		Germany	393 CD	399	942 CD; 375 trios	1082
2007	Perlegen 165k	Raelson et al. [66]	CD		Canada (Québec founder population); Germany	382 CD		752 CD; 521 trios	828
2007	Illumina 300k	Libioule et al. [67]	CD		Belgium; France	547 CD	928	1266 HC	559
2007	Affymetrix 500k	WTCCC [68]	CD		UK	1748 CD	2938	1182 CD	2024
2008	Meta-analysis	Barrett et al. [1]	CD		UK/North/America/France/Belgium	3230 CD	4829	2325 CD; 1339 trios	1809
2008	Nonsynonymous SNP 14k	Fisher et al. [69]	UC		UK	905 UC	1465	2028 UC	3029
2008	Affymetrix 500k	Franke et al. [27]	UC		Germany	1167 UC	777	1855 UC	3091
2008	Illumina 550k	Kugathasan et al. [70]	IBD	Early-onset	North America; Italy	1011 IBD	4250	173 IBD	3481
2009	Illumina 550k and 300k	Silverberg et al. [71]	UC	No proctitis	North America	1052 UC	2571	1405 UC	1115
2009	Illumina 550k; Affymetrix custom 10k	Asano et al. [72]	UC		Japan	749 UC	2031	635 UC	1026
2009	Affymetrix 1000k	UKIBDGC [73]	UC		UK	2361 UC	5417	2321 UC	4818
2009	Illumina 550k	Imielinski et al. [74]	IBD	Early-onset	North America; Scotland; Italy	1636 CD; 724 UC	6158	829 CD; 120 UC	5805
2010	Affymetrix 1000k	Franke et al. [75]	UC		Germany; UK; Belgium; Norway; Greece; Baltic Countries	1043 UC	1703	2539 UC	5428
2010	Illumina 550k	Wang et al. [76]	IBD, T1D	Overlap analysis	European ancestry	1689 CD; 777 UC; 989 T1D	6197		
2010	Meta-analysis	McGovern et al. [77]	UC		North America/Sweden meta-analysis; Italy; Netherlands	2693 UC	6791	2009 UC	1580
2010	Meta-analysis	Franke et al. [2]	CD		European ancestry	6,333 CD	15,056	15,694 CD; 414 trios	14,026
2011	Meta-analysis	Anderson et al. [3]	UC		European ancestry	6,687 UC	19,718	9,628 CD	12,917
2012	Meta-analysis (GWAS and Immunochip data)	Jostins et al. [2]	IBD		European ancestry	5,937 CD; 6,945 UC	21,770	14,763 CD; 10,920 UC	15,977
2012	Illumina 300k/550k; Affymetrix 500k/6.0	Kenny et al. [78]	CD		Ashkenazi Jews	907	2345	971	2124
2013	Illumina Quad610	Julia et al. [79]	CD		South Europe	1341	1518	1365	1396
2013	Illumina HumanHap550	Yamazaki et al. [80]	CD		Japan	372	3389	1151	15800
2014	Illumina OmniExpress/Omn1-Quad arrays	Yang et al. [81]	CD		Korea	533	800	1779	1709
2014	Illumina Quad610	Julia et al. [82]	UC		South Europe	825	1525	1073	1279

Abbreviations: CD = Crohn's disease; GWAS = Genome-wide association studies; IBD = Inflammatory bowel disease; UC = Ulcerative colitis.

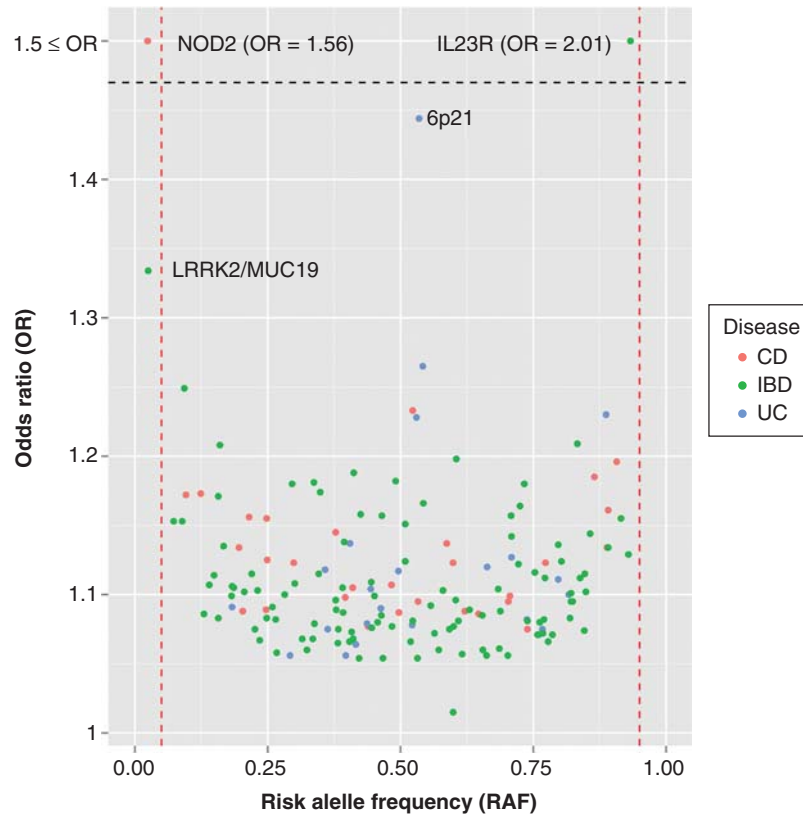


Figure 1. Risk allele frequency versus odds ratio plot. Data for 163 index SNPs were extracted from the latest meta-analysis on inflammatory bowel disease (IBD) [2]. The risk allele frequency is plotted on the X-axis and shows that most identified alleles have a minor allele frequency $>5\%$. Alleles can be protective (risk allele frequency >0.50) or exert risk (<0.50). The Y-axis shows the odds ratio and that almost all identified risk variants have an odds ratio <1.30 (corresponding to >0.77 for protective variants). Colors indicate whether the particular variant reached genome-wide significance for Crohn's disease (CD) (red), ulcerative colitis (UC) (blue) or both diseases (green). The by far strongest UC-associated locus was the noncoding SNP rs6927022 ($p = 4.71 \times 10^{-133}$; OR [odds ratio] = 1.44; 95% CI = (1.39–1.50)) near the class I gene *HLA-DQA1*. However, more relevant UC-associated *HLA-DRB1* alleles as *DRB1*15:01*, *DRB1*07:01* and *DRB1*01:03* show much larger effect sizes [20].

order to reach sufficient statistical evidence for a robust variant versus disease association. The currently accepted significance threshold is a p -value smaller than 5×10^{-8} , under the assumption that appropriate and adjusted (ancestry, gender, other covariates of importance) analyses have been carried out, a threshold that is also known as 'genome-wide significance' [4]. It has been assumed that all risk loci with a minor allele frequency $>5\%$ in the general population and an OR >1.2 have been identified to date in IBD patients with European ancestry by means of GWAS [5]. One clearly has to distinguish between common and rare susceptibility variants. Common variants are more frequent than 1% in the general population. Current GWAS are usually designed with enough statistical power to detect such variants [6]. Variants with $<1\%$ frequency are generally referred to as rare variants and variants that are observed only within single pedigrees are termed private variants (see e.g. Table I in Cirulli and Goldstein for variant classification [7]). If such private variants have arisen *de novo*, that is, the

nonreference allele is absent in the parents but present in the child's genome, this variant can be referred to as a true mutation. Since sequencing approaches identify common, rare, private and *de novo* variants at once (in contrast to microarrays where known single nucleotide polymorphisms [SNPs] are tested) these variants are collectively summarized as single nucleotide variants (SNVs). More than 163 independent autosomal genetic risk loci have been identified to this end for IBD, specifically for the two major IBD subtypes CD and UC. The disease variances explained by these 163 loci, actually comprising 193 statistically independent SNPs, are 13.6% and 7.5% for CD and UC, respectively [8]. Twin and family studies for IBD have shown that a child has a 26-fold increased risk for developing CD when another sibling already has CD and the risk is increased 9-fold in the case of UC [9]. The afore mentioned facts not only underline the importance of genetic factors but also illustrate that there are other contributing effects that need to be taken into account and that require future investigations.

Disease-specific versus shared associations

On the genetic level, 110 out of 163 loci are shared between CD and UC [8], even when taking effect directions into account and not only positional overlap (which is actually common practice in many studies to show a shared etiology). This molecular similarity between CD and UC perhaps reflects the clinical similarities observed by the gastroenterologist between both diseases. It further indicates a heterogeneous and continuous disease spectrum and may explain why clinicians often fail to make a definitive diagnosis and why for a large fraction of patients the diagnosis changes during disease course. Given that IBD is an immune-mediated disease, it is not surprising that several disease variants cluster in pathways relevant for the innate (e.g. *NOD2*, *ATG16L1*) and adaptive immune system (*IL23R*, human leukocyte antigen [HLA] locus). Moreover, >50% of the known IBD loci overlap with those of other immune-mediated diseases. The strongest overlaps to date have been observed between IBD and ankylosing spondylitis, psoriasis, and primary sclerosing cholangitis. The following review articles address the shared etiology of IBD and related diseases in detail [10,11]. The observation that distinct diseases affecting different organs (e.g. intestine vs. joints vs. skin) have similar genetic etiologies suggests two potentially fruitful future directions: first, clinicians working in different disciplines such as gastroenterology and dermatology should interact even more not only in research but also on the clinical level (e.g. treatments for disease A might also work in disease B, joint discussions about ‘complicated patients’ in analogy to cancerous diseases and comprehensive cancer centers). Second, systematic genetic cross-phenotype association studies should be carried out in order to dissect the disease associations even further and to identify additional associations, also of pleiotropic nature. It is not yet clear for most joint loci it is not yet clear whether (a) the same variants are associated and if (b) effect directions are similar. And while it is easier to identify joint effects, it remains a challenge to annotate a given variant as disease-specific because confidence intervals (CI) of OR are often overlapping. Therefore, studies reporting disease-specific associations should be handled with caution and replication studies or rigorous thresholds are required for claiming a disease-specific finding. Often, patients also have more than one disease and it will be a challenge to account for comorbidities – while in some instances the secondary disease is yet unknown because most cases are not followed up longitudinally – in association studies.

Further fine mapping is needed

Although the costs for sequencing complete human exomes (the entirety of all known human exons) and even genomes have significantly dropped in the last years, large-scale resequencing studies for IBD, to an extent as they were carried out using SNP arrays, are still scarce and work in progress. Only a few targeted sequencing efforts, most of them employing DNA pooling approaches, have been carried out for IBD [12–14]. Therefore, a comprehensive list of rare and also private variants relevant for IBD will still require several years of research. To this end, for most GWAS loci it cannot be concluded that an associated SNP is actually related to the disease on a biological level, that is, being the actual causative variant, or whether the index SNP is just a marker for a neighboring causative variant. Functional experiments are required to ultimately prove the disease-relevance of a particular candidate gene or risk variant. Most of the associated genomic regions contain more than one gene (~1444 candidate genes before and 300 after *in silico* prioritization in Jostins et al. study [8]) and some do not even contain any known coding sequence. Notably, twice as many of the IBD susceptibility variants are predicted to impact gene expression (so-called Expression quantitative trait loci [eQTL] association) compared to index variants that are coding or in linkage disequilibrium with a known coding variant. Functional experiments are even more important, or rather essential, for private variants, as classical statistical approaches to prove disease-relevance are not applicable to singular observations. Besides, from a biological and also statistical point of view, more than one variant in a given risk locus can independently contribute to disease risk. However, many studies report only an ‘index SNP’ with the smallest *p*-value for the identified genetic loci. To address this shortcoming, concerted fine mapping efforts are on the way for the known genetic risk loci employing large and dense array data sets. Similar studies have been completed for other diseases, such as psoriasis [15], and have shown an additional layer of complexity and allelic heterogeneity at known risk loci. In addition, more transethnic genetic studies are needed for IBD in order to explore the genetic architecture of IBD risk across different populations [16].

Role of genetic variants in the HLA complex

The highly polymorphic genomic region on chromosome 6p21 from megabases 25 to 34 is known as the major histocompatibility complex (MHC) or as the

HLA region. It contains many genes that are relevant to immune system function. Pinpointing the causative variants and even the exact association signals in the HLA remains a challenge.

The by far strongest UC-associated locus in the largest genetic study for IBD to date [8] was SNP rs6927022 ($p = 4.71 \times 10^{-133}$; OR = 1.44; 95% CI = (1.39–1.50)) near the class I gene *HLA-DQA1*. The association signal in the HLA region for CD proved to be strongest for SNP rs9264942 ($p = 4.96 \times 10^{-28}$; OR = 1.15; 95% CI = (1.11–1.18)) located in the *HLA-B* gene locus of the MHC class I region. Even linkage studies, which were carried out over 10 years ago, identified the HLA-locus as a genome-wide significant genetic risk locus for IBD [17], formerly known as the *IBD3* locus. Much progress has been made since this linkage era in mapping the IBD-relevant signals within the HLA, one of the most polymorphic regions in the genome with extensive linkage disequilibrium and a high gene density. Furthermore, the HLA is an ancestry-informative locus and diverse across different ethnicities. Several association scenarios may exist and they may be different in CD and in UC: (1) certain class I alleles may be associated, (2) class II alleles may play the prominent role, (3) association signals may reside outside the classical class I and II loci, or (4) a combination of (1)–(3) may be true. Recent decrease in cost for HLA typing – and more prominently imputing classical HLA alleles [18,19] – have facilitated large-scale *in silico* HLA fine mapping approaches using existing dense GWAS SNP array data. The latest fine mapping study of the IBD association signal on 6p21 was carried out by Duerr and colleagues [20]. They analyzed 562 UC patients, 611 CD patients and 1428 control individuals analyzed for 10,347 SNPs within the HLA region (29.299–33.884 Mb; National Center for Biotechnology Information [NCBI] build hg18). While only suggestive association evidence was obtained for CD (rs17880124; $p = 3.82 \times 10^{-5}$; OR = 2.23; 95% CI = (1.52–3.27) for the G allele), the best UC association signal was narrowed down to the single amino acid position 11 of the HLA-DRB1 P6 antigen-binding pocket. The best SNP signal was observed for rs2647025 ($p = 1.94 \times 10^{-12}$; OR = 1.95 for the G allele; 95% CI = (1.62–2.35)), which was in moderate linkage disequilibrium (LD) ($r^2 = 0.63$) with the previously reported SNP rs9268853 of the IIBDGC UC GWAS meta-analysis [3]. Including classical HLA alleles, allele groups and binary HLA amino acid information in the analyses with the SNP data, resulted in rs9269955 as the most significant association with UC in the HLA ($p = 2.67 \times 10^{-13}$; OR = 0.51; 95% CI = (0.43–0.61) for allele C). rs9269955 determines the codon for the aforementioned amino acid position

11 where six different amino acid alleles are observed in the population. The relevant UC-associated *HLA-DRB1* alleles were *DRB1*15:01*, *DRB1*07:01* and *DRB1*01:03* with the latter having the largest effect size of 38.39 (95% CI: 7.50–169.60).

Given that the association of variants within the HLA region with both UC and CD are very strong with large effect sizes – and given the particular importance of this region in immunity in general – the understanding of IBD pathophysiology will only be accomplished by understanding the causative nature of the HLA association in more detail.

For CD, HLA associations have been reported in several studies (e.g. SNP rs1799964 in Franke et al. CD meta-analysis [2]); however, large coordinated fine mapping efforts employing more than 60,000 samples are on their way to accurately map signals and to compare the association signal to UC in more detail (Goyette and Boucher et al., submitted). However, array-based mapping approaches have a limited resolution and will likely not provide answers to all open questions concerning the HLA association. The identification of a potentially existing antigen that contributes to the outbreak of IBD will ultimately require synergistic and complementary approaches such as, for example, T-/B-cell, repertoire sequencing and peptide elutions with subsequent analyses [21,22].

Clinical relevance of genetic findings

One remarkable observation is that even though many individuals carry risk alleles, they do not develop one of the diseases (see Figure 2 in Festen et al. [23]). It may be of interest to stratify healthy individuals and patients by their cumulative genetic risk. For example, individuals from the extreme tails of the risk score distribution can be selected and analyzed in more detail (e.g. run stratified expression analyses). Is a healthy person with an extremely high IBD risk score maybe not healthy, that is, has other symptoms? Is this group maybe having subclinical disease symptoms? Is there a strong environmental factor, such as a chronic infection with a pathogen, in patients with an extremely low genetic risk (and without any ‘attractive’ rare/private variants)? However, it has to be kept in mind that most of these risk alleles only increase the total risk by a modest amount. Furthermore, even if an individual has a high cumulative risk score, the disease might not develop. Potential explanations are that it does not only matter how many risk alleles an individual carries but also what kind of combinations of risk alleles are present. Furthermore, the lifestyle and

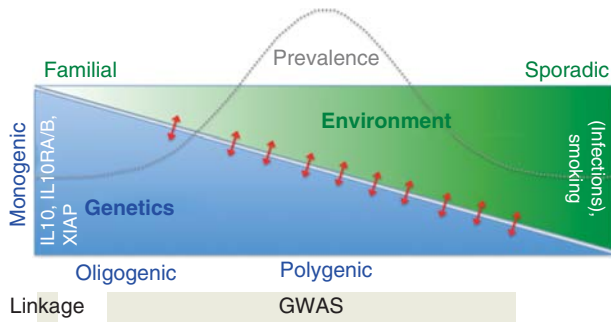


Figure 2. Inflammatory bowel disease (IBD) etiology as an example for a complex chronic inflammatory disease. In most patients (prevalence illustrated by dashed curve), IBD develops through (mostly unknown) environmental factors – also including the gut microbiome – that act on a genetically predisposed host. Yet, cases have been described where highly penetrant genetic mutations cause (monogenic) IBD. It is still discussed how relevant infections are in the onset of IBD. The gene–environment interactions (red arrows) are important in disease etiology and should be studied in more detail in the future. While genome-wide association studies (GWAS) are successful to identify common disease loci, linkage studies are more powerful to identify monogenic forms of IBD. Recently, exome-sequencing has been employed for these early-onset forms since next-generation sequencing provides single base pair resolution for mutation detection and also allows for *de novo* mutation detection. Modified from Kaser *et al.* [83].

the exposition to other environmental factors seem to play the most important role.

On monogenic and oligogenic forms of IBD

While IBD is regarded as a genetically complex disease with a large number of variants, each contributing a small part of the total heritability, monogenic and oligogenic forms of IBD also exist (see left part of Figure 2). While the disease onset for CD as well as UC can occur at any age, the peak incidence is in early adulthood. Monogenic and oligogenic forms, however, usually manifest earlier in life. Early-onset cases of IBD, with a disease manifestation during the first 18 years of life, also often show a more severe disease course with a higher risk of complications, very-early-onset cases even develop the disease during the first 8 years of life. GWAS and candidate gene sequencing studies performed specifically for early-onset IBD have identified several genes and variants that seem to be exclusively associated with this form of IBD. Nevertheless, there is significant overlap between the genetic causes of early- and adult-onset IBD. A large spectrum of monogenic diseases can also present with IBD-like intestinal inflammation [24]. The underlying causes include defects of the intestinal barrier function, immunodeficiencies with a disturbed granulocyte and phagocyte activity, hyper- as well as auto-inflammatory effects and disturbed B- and T-lymphocyte function.

Several studies have identified shared genetic factors underlying monogenic or early-onset and adult-onset IBD cases with rather oligogenic or polygenic causes. A genetic linkage analysis followed by candidate gene sequencing in two consanguineous families with children affected by early-onset IBD during the first years of life revealed homozygous mutations in genes for the IL10 receptor (IL10R) subunit proteins [25] to be causative. In two other patients, who did not show variants in the *IL10RA/B* genes, distinct mutations in the *IL10* gene itself were shown to be disease-causing [26]. *IL10* was also associated with adult-onset UC [27] and CD [2] in GWAS. An autosomal-recessive case of inflammatory skin and bowel disease in two siblings with consanguineous parents was shown to result from a homozygous deletion in *ADAM17* (ADAM metalloproteinase domain 17) by using a combination of SNP-homozygosity mapping and targeted NGS-Sequencing of the linkage regions [28]. Although the direct overlap between key genes that have been associated with IBD and IBD-like monogenic disorders is surprisingly low [24], the affected proteins often interact directly or indirectly with each other and share common signaling cascades that contribute to IBD etiology. Exome sequencing of a male patient with a severe case of CD during the first year of life recently revealed a *de novo* nonsense variant in the X-linked inhibitor of apoptosis gene [29]. Targeted sequencing in ~1,300 CD patients then revealed two additional missense and one nonsense variant each private to patients that showed an age of onset between 13 and 16. Variants in one gene can thereby have a varying impact on disease manifestation and the same gene can simultaneously be involved in monogenic early-onset as well as oligo- or polygenic adult-onset cases. The analyses of monogenic forms of IBD therefore have the potential to give important insights into further mechanisms contributing to the common forms of IBD. For more details on early-onset monogenic IBD, we recommend reading the reviews by Uhlig and colleagues [24,30].

From genetics to mechanistic insights

Current therapy of IBD symptoms cannot prevent the need for surgery in more than half of the patients. Therefore, novel treatment strategies and a detailed understanding of the disease causes are of high importance. Learning about the relevant biological pathways and proteins that are influenced by the genetic susceptibility variants may help in defining new targeted therapies and the pathways may also point at important environmental risk factors that should be studied. Several excellent review articles have summarized the current understanding of the

pathophysiology of IBD and how the identified genetic susceptibility factors may contribute to disease etiology. Perhaps one of the most exhaustive review articles, which we recommend reading for understanding IBD pathophysiology, has been published by Kaser et al. [31]. Several other recent reviews have elucidated the mechanistic insights and pathways gained through genetic studies [6,32]. One should be cautious though in interpreting genetic association loci as most of these loci, contain more than one candidate gene as mentioned previously (in contrast to the monogenic forms where interpretation of loci is much more straightforward). Therefore, well-designed mechanistic studies are needed to prove the disease-relevance of a particular candidate gene/variant in a given GWAS locus. Of course, *in silico* prioritization approaches may be useful to narrow down the list of potential candidate genes in a GWAS locus; however, although these approaches may be very accurate, results should be handled with care until the functional proof is ultimately available. Often, candidate genes are also (cherry-) picked based on the previous experience and knowledge of the interpreting scientist. Only a handful of candidate genes and pathways exist for IBD where there is no doubt that these are playing a causative role in disease etiology, for example *NOD2*, *IL10*, *IL23R* and *ATG16L1*. The latter is an excellent example how genetic findings fuel subsequent functional studies. After our initial discovery of a disease-associated coding variant in *ATG16L1* in 2007 [33] – and hence the first implication of the autophagy pathway in IBD disease etiology – several high-impact articles were published that elucidated the role of *ATG16L1* in IBD in more detail, also employing animal models [34–40]. While no publication mentioned the gene *ATG16L1* in its abstract before the initial discovery in 2007, more than 281 publications have listed it afterward (until November 2, 2014). Out of these 281 publications, 203 abstracts also mention the disease terms CD or IBD (PubMed query: ‘*ATG16L1* AND [‘Crohn’ OR ‘Crohn’s’ OR ‘inflammatory bowel disease’ OR ‘inflammatory bowel diseases’]). Another 177 independent manuscripts mentioned the word ‘autophagy’ and CD/IBD in their manuscript. Therefore, we feel that genetic association studies were a worthwhile undertaking, in particular in IBD genetics research. GWAS point to regions, candidate genes, functional elements or pathways that should be studied in greater depth. In addition, we have learnt that the complex diseases CD and UC are much more than different diseases with an aberrant T-cell response (past view of IBD pathophysiology), but rather very related diseases with several disturbed cellular pathways (autophagy, ER stress response,

innate immunity, epithelial barrier dysfunction, ...; see Figure 2 in [6] for details). Exome sequencing for cases with monogenic IBD forms nicely complements the GWAS approach and points to additional or similar pathways as identified in GWAS.

Conclusion and outlook

Although the genetic risk map for IBD has grown tremendously in the last decade, the exact disease etiology and disease cause remain largely unknown. One of the great challenges of future IBD research is to understand how the disease-associated mutations actually contribute to disease etiology. When it comes to predicting disease or clinical subphenotypes, larger studies have been published, showing that this is not sufficiently accurate enough for clinical purposes as most of the identified variants are common and have low penetrance [41,42]. Therefore, and due to the large environmental component in disease development, it is likely that genetics alone won’t succeed in predicting disease outbreak and course, but that the genetic risk maps will serve as an important puzzle piece in a statistical model that considers nongenetic data in parallel. For rare, severe early-onset forms of IBD the situation is different as compared to the more common and complex disease forms (Figure 2). In patients presenting with severe, early-onset IBD, a single rare/private variant or a mutation may be the only cause; however, incomplete penetrance and allelic heterogeneity may also blur predictions in these cases. Nevertheless, diagnostic candidate gene resequencing should be carried out in these early-onset patients in clinical practice.

The large number of genetic findings demands for large-scale functional genomics analyses and cannot only rely on mouse models. However, introducing specific and multiple mutations into the mouse genome has become easier and quicker with the advent of the clustered regularly interspaced short palindromic repeats (CRISPR)/Cas engineering technique [43].

According to the ENCODE project [44] most disease loci overlap with regulatory regions [45]. It is likely that these loci in noncoding regions play a regulatory role and influence the activation and concentration of proteins that are relevant to immune system function and behavior. It may be a greater challenge to characterize such regulatory variants, however, first attempts have been made, as in the case of *PRDM1* [46]. The common CD risk allele C at SNP rs7746082, identified in GWAS, correlates with reduced expression of *PRDM1* in ileal biopsy specimens and peripheral blood mononuclear cells. Notably, exome sequencing may not identify the relevant susceptibility variants in most patients as >84% of the

IBD-associated variants are not located in exons (~16% are nonsynonymous variants) and as more than ~36% are eQTLs, thus affecting gene expression [5]. This fact plus that larger genetic variants – such as copy number variants, insertions/deletions, structural variation and mobile element insertions – are understudied in IBD genetics research (the large CD-associated deletion upstream of the *IRGM* gene being one of the few IBD CNV success stories [47]), emphasizes that large-scale whole genome sequencing efforts are urgently needed. Moreover, larger gene expression data sets for the disease-relevant tissues and cell types should be generated to allow for high-resolution and more eQTL studies since eQTLs are often cell-specific [48].

The aforementioned ongoing and future genetic studies in IBD will further increase the resolution of the IBD genetic risk map. However, the known phenomenon of the ‘missing heritability’ will likely remain after the mapping efforts [49]. Therefore, other opportunities to identify heritable components should be considered [50]. One opportunity could be to study epigenetic factors that may be transmitted transgenerationally. A review from Jirtle and Skinner provides a good overview on this topic [51]. More recently, several studies in mice presented evidence that different environmental exposures can lead to epigenetic changes in the next generations [52]. Especially the exposed sperm epigenome of male mice has received a lot of attention recently [53–55]; in accordance with a zebra fish study [56]. Human studies exist that suggest an important role of the parental epigenome ‘recording’ environmental exposures and a subsequent inheritance of this ‘recorded information’ to the next generation (mostly *in utero* environmental exposure that affects F1 generation) [53]. Only if an effect is seen in the F3 generation, since an exposure in a pregnant woman may directly affect the F1 and F2 generation, one can speak of a true transgenerational effect. In the paternal model, a phenotype change must be also seen in the F2 generation. Therefore, it may be fruitful to study whether certain environmental factors can induce epigenetic changes that are heritable and that could further explain the familial occurrence of IBD. Moreover, deleterious epigenetic changes may be solely responsible for disease onset in some patients. The picture becomes more complex if one considers that genetic susceptibility factors make a host susceptible to certain environmental exposures, which in turn may influence the epigenomes of cells and that these epigenetic changes could be inherited to offspring cells or children. Interestingly a recent study found that exposure to DDT resulted in obesity in half of the rats in the F3

generation (every second rat had obesity). No effect of the epimutations was observed in the F1 and F2 generation, showing that multigenerational studies are necessary to study transgenerational epigenetic inheritance [57].

It has been hypothesized that host–bacteria interactions in the gastrointestinal tract trigger chronic inflammation in the genetically susceptible host. In fact, several studies indicate that IBD patients have a dysbiosis of the intestinal microbiota [58]. Protective and risk-associated bacterial genera were further identified and transplantation/co-housing experiments in mice even point towards a disease-causing (and not only disease-correlated) role of the microbiota [59]. It has been speculated – and not yet convincingly demonstrated – that the genetic IBD association signals are enriched for genes that are relevant for host–microbiota interaction in the intestine [60]. Jostins *et al.* have also carried out preliminary selection analyses for the 163 IBD loci, which shows a remarkable enrichment of evolutionary signatures in the IBD loci. Since the host and the microbiota have coevolved, genetic data from patients should be jointly analyzed with gut microbiome data from the same patients and proper association/interaction tests be carried out. First candidate gene studies (e.g. for *FUT2* and *IRGM*) in humans indicate that genome-wide host–microbiota association studies may be as fruitful as GWAS [61,62].

In addition, more information about the lifestyle of patients could prove to be useful. More importantly, large longitudinal cohort studies are needed to ‘follow individuals into disease.’ Individuals with subclinical disease may be the ideal study cohort to identify predictive biomarkers and also the actual factors that will drive the development of clinical symptoms. Once the disease is chronic, it may be too late to change certain triggers, leaving only the option to treat clinical symptoms, as is the current state-of-the-art in clinical practice (anti-inflammatory drugs).

Big and multilevel omics data, including large genomic resequencing and genetic data, steadily increase for IBD and its related diseases. Concerted and interdisciplinary efforts of, for example, analysts, clinicians and molecular biologists are needed to identify meaningful results from the data deluge. We encourage all researchers to share their data and methods as open as possible in order to comprehensively understand this complex disease and we look forward to (contribute to) the next 10 years of IBD (genetics) research.

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